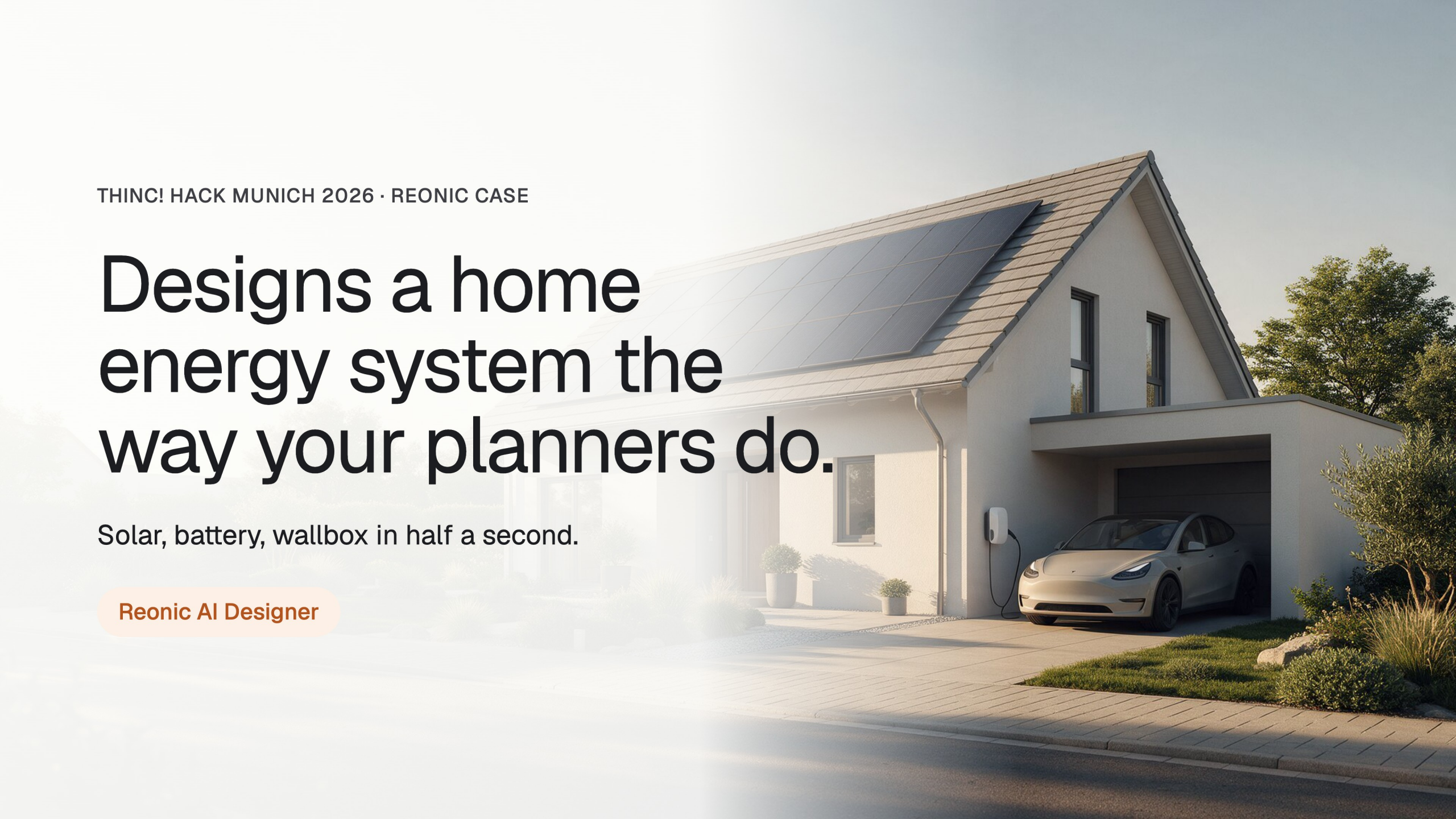
THINC! HACK MUNICH 2026 · REONIC CASE

# Designs a home energy system the way your planners do.

Solar, battery, wallbox in half a second.

Reonic AI Designer





THE PROBLEM

# Planners “spend hours on every single design”, and “the first credible offer often wins”.

The answer already sits in the installer’s past offers.

WHAT WE BUILT

# One engine. Three doors.

1 · Voice call or phone form

2 · Planner

3 · Homeowner link

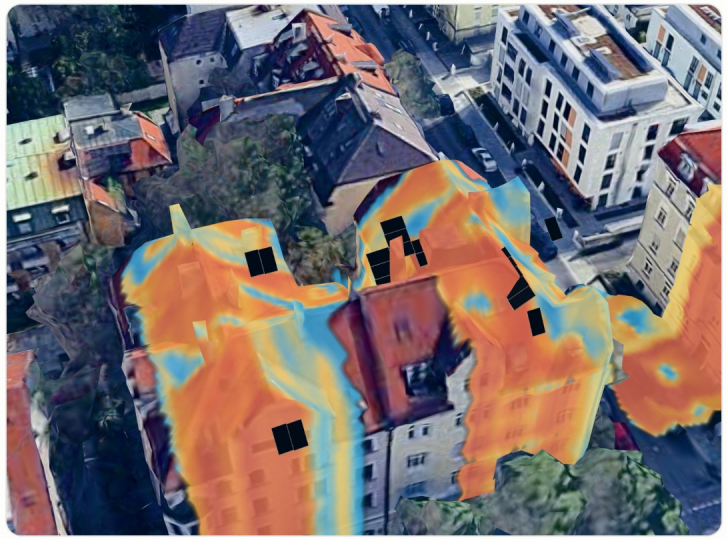


THE PRODUCT

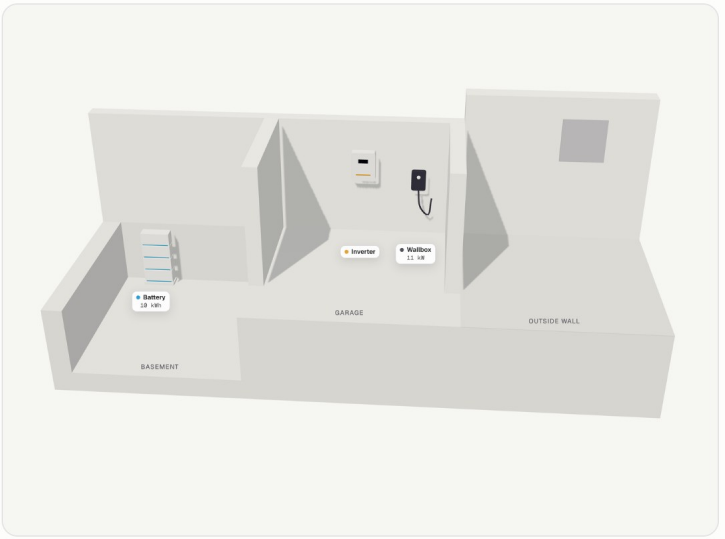
# Their house, not a template.




**Their real street**



**Panels where the sun is**




**Floor plan → rooms**

315 HOMES IT NEVER TRAINED ON

# 24 % less solar error than the best rule of thumb.

**2.93 vs 3.85 kWp**

Solar error per home, all 315 homes

**77 % vs 6 %**

Add-on sales: no solar where none was sold

Compared with the best of three rules of thumb · live on /eval

HONEST NUMBERS

# Where it wins, and where it doesn’t.

## Better

| | |
|---|---|
| New systems, size | 2.45 vs 2.64 kWp |
| Expert inside our range | 82 % vs target 80 |
| Expert near 1 of our 3 | 66 % vs – |
| Battery size | 2.42 vs 2.71 kWh |

## Not better

| | |
|---|---|
| Within ±1.5 kWp | 40 % vs 43 % |
| Battery yes/no | 88 % vs 90 % |
| Wallbox yes/no | 72 % vs 73 % |

NOT A BLACK BOX

# Every size comes with its reasons.

**8.1–10.4**

Range, not a guess

**+1.0 kWp**

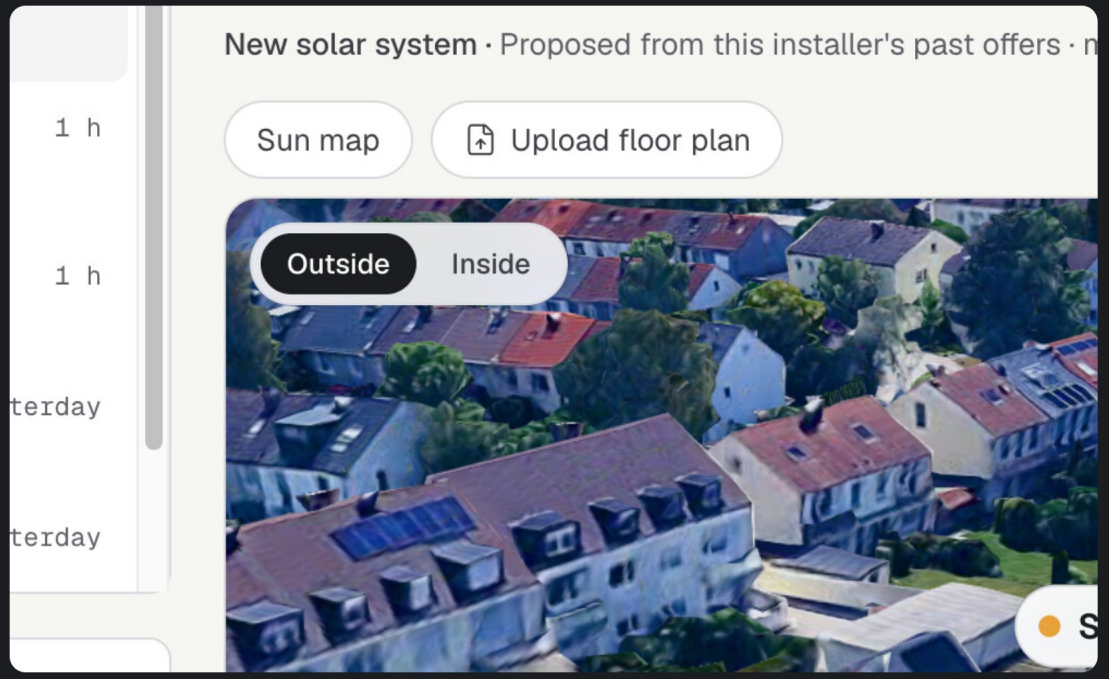
“Heating demand”: the why

**5 homes**

Similar ones they built

NEXT

# Real street today. Drone-sharp next.




**Live today**



**Concept, with high-res capture**

MEASURED ON PRODUCTION

# From phone call to homeowner link in 29 seconds.

**29 s**

Call start → homeowner link

**4.6 s**

Call end → shared offer

**0.5 s**

Full design from the engine

# One engine for the call, the planner and the homeowner.

Every size explained. 24 % less solar error. Homes it never saw.

Reonic AI Designer

HOW IT DECIDES

# One chain, not three guesses: each decision sees the one before it.

### Heat pump

A rule, offered as a customer option for gas and oil homes. Heat demand ÷ 2,450 h, snapped to the sizes the installer sold. Never imposed.

→

### PV yes/no, then size

Sees the total demand incl. EV and heat pump. Gradient-boosted trees with a robust loss; more demand or an EV can never shrink the PV. P10–P90 range, conformally calibrated.

→

### Battery yes/no, then size

Sees the planned PV. Size = the classifier's pick blended with the installer's own kWh-per-kWp rule, snapped to its catalogue.

→

### Wallbox

EV → wallbox. The simple rule beat the model on training data, so the engine uses the rule and says so.

### Before the chain: new system or add-on?

Battery retrofit, wallbox-only or new PV is its own decision. The question "Solar already on the roof?" is its strongest input, so the call and the form ask it first.

### Learned from

1,994 offer options across 1,593 signed projects from two installers, parsed from the offers' line items, 20 projects hand-checked. Decisions learned from each project's first full design, sizes from every option: Reonic confirmed each option is a valid design. Every size snaps to the installer's catalogue.

### Physics on top

An hourly year with PVGIS yield per roof tilt and orientation gives yield, self-consumption and autarky, and flags what is unusual for these installers.

WHAT THE DATA TAUGHT US

# Four things experts do that no textbook rule says.

## new vs. add-on

**The decision that matters is not the kWp**

Recognising battery retrofits and wallbox-only orders from the planner's inputs cuts the whole-project solar error by 24 %. Pure size error improves only from 2.64 to 2.45 kWp.

## 1 input, 1 question

**Yearly consumption does most of the work**

Installer + consumption alone already reach the band-median rule; removing consumption costs 0.9 kWp. The question "solar already on the roof?" would be worth the most: as a what-if, 2.93 → 2.15 kWp over all homes.

## 2.77 vs 2.49 kWp

**Experts do not size for maximum return**

Sizing for the economic optimum is worse than learning from them. Installers size PV at about 2 kWp per MWh, and the battery by the PV size, not by consumption: 2.25 vs 4.46 kWh error for the "1 kWh per 1,000 kWh" rule.

## 7–8 kWp + battery

**Small homes get a full system on purpose**

New systems for homes using 1,500–2,500 kWh got a median 7.2 kWp from the experts (n = 109). Textbook plausibility rules would flag that as oversized, so our checks are calibrated on expert practice, and the engine proposes the same.

IN THREE NUMBERS

## 0.5 s

### A complete design

PV, battery, wallbox, heat-pump option. Every size with its reasons, range and alternatives.

## 24 %

### Less solar error than the best rule of thumb

Because it tells a new system from an add-on, the way the installer's own planners do.

## 315

### Homes it never saw

Held out by customer. Every number in this deck is measured there, including where we are only on par.

One engine behind a phone call, a planner workspace and a homeowner link. The engine decides every size; an LLM never picks one.